

Quantum-Inspired Anomaly Detection for Memory-Based Intrusion Detection Systems



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Abstract

Memory-based cyberattacks are one of the most difficult threat classes to detect. They operate entirely inside volatile memory. Nothing is written to disk. No file is created or modified. Signature-based antivirus tools scan files and match known patterns. They cannot detect an attack that leaves no file trace. This gap is well documented and widely exploited. Heap spraying defeats Linux memory randomisation at a 97.92% success rate using timing side-channels. Process injection through the Linux `/proc` filesystem reads live process memory without creating any new process. Memory-resident malware evades all traditional detection methods. Yet most organisations still rely on signature-based tools as their primary defence.

MemoryWatch is a host-based intrusion detection system designed to close this gap. It monitors process memory behaviour in real time using syscall frequency and `/proc` access patterns as detection signals. The system targets three specific attack types: heap-based buffer exploitation, unauthorised process memory access through `/proc` and memory-resident malware. These were chosen because they are real, actively exploited and currently undetectable by mainstream security tools.

The classical detection layer uses an Isolation Forest model trained unsupervised on normal behaviour data from the UNSW-NB15 dataset. A dynamic threshold computed per process class flags anomalies without relying on a fixed global value. This approach reduces false positives while maintaining high recall. The quantum-inspired layer implements a ZZFeatureMap fidelity kernel using Qiskit's QuantumKernel API. A one-class SVM uses the kernel matrix for anomaly scoring. Both layers are trained and evaluated on the same preprocessed UNSW-NB15 data. Results are compared directly using false positive rate, recall, F1 and AUROC as primary metrics. Accuracy alone is not used because it is a misleading measure for imbalanced intrusion detection datasets.

The quantum layer is explicitly a research contribution and not a production component. Current quantum hardware operates in the NISQ era with high gate error rates, limited qubit counts and short coherence times. Training was limited to 300 samples due to circuit complexity constraints. All quantum results come from Qiskit statevector simulation and reflect detection quality only. Simulation speed cannot be taken as evidence of hardware deployment performance.

The results show that Isolation Forest outperforms the quantum kernel model on binary detection accuracy and training scalability. The quantum model shows a marginal improvement in AUROC on multiclass near-boundary cases. This is consistent with findings in the literature that quantum advantage in IDS is most visible in multiclass feature-space separation rather than binary classification. Every design decision in MemoryWatch maps directly to a finding in the literature. The system is practical, measurable and honest about the current limits of quantum deployment.

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1 Introduction

1.1 Motivation

When a program runs, sensitive data sits in RAM as plain text. Passwords, encryption keys and session tokens all live in volatile memory during execution. The CPU needs direct access to this data. That is also what makes it a target.

Attackers who can read live process memory extract credentials silently. Nothing is written to disk. No file is touched. Signature-based antivirus tools scan files and match known patterns. They cannot detect an attack that only exists in RAM.

This is not a theoretical concern. Lee et al. (2023) showed that heap spraying with a timing side-channel defeats Linux memory randomisation 97.92% of the time across ten real kernel vulnerabilities. Nasereddin and Al-Qassas (2024) confirmed that process injection through the Linux `/proc` filesystem is completely fileless. Dehfouli and Habibi Lashkari (2025) reviewed the full landscape of memory-resident malware and found that traditional detection methods fail against this threat class without exception.

Most organisations still rely on signature-based tools as their primary defence. The gap between what these tools detect and what attackers use is wide.

1.2 Context of the Project

MemoryWatch was developed as an Action Learning project at EPITA Graduate School of Computer Science. It sits at the intersection of three areas: host-based intrusion detection, classical machine learning and quantum-inspired computing. Each area had a piece of the answer. No existing system joined them into a single pipeline targeting memory-based attacks.

The project is deliberately scoped. The goal is not to rebuild a general-purpose IDS. The target is the specific class of attacks where existing tools fail completely attacks that live only in volatile memory and leave no file trace.

1.3 Objectives

The project sets out to achieve four things.

Build a host-based IDS that monitors process memory behaviour in real time using syscall frequency and `/proc` access patterns as detection signals.

Train an Isolation Forest model on normal behaviour data from UNSW-NB15 and evaluate it using false positive rate, recall, F1 and AUROC.

Implement a quantum-inspired classification layer using Qiskit's ZZFeatureMap kernel on the same dataset and compare it directly against the classical baseline.

Produce an honest comparative analysis that separates benchmark claims from real-world deployment claims and discusses quantum limitations explicitly.

1.4 Overview of the Thesis

Chapter 2 reviews the research across four areas: memory attack techniques, machine learning for anomaly detection, dataset quality and evaluation metrics and quantum machine learning for cybersecurity. Chapter 3 draws this into a state-of-the-art analysis and identifies the gap MemoryWatch fills. Chapter 4 describes the system architecture. Chapter 5 covers the methodology. Chapter 6 reports results. Chapter 7 concludes and identifies directions for future work.

2 Literature Review

The research for this thesis was divided among the four team members along the project's main pillars. Vignesh focused on memory attack techniques and the threat model. Ghita covered the classical machine learning pipeline and dataset validation. Prachin handled evaluation metrics and dataset limitations. Esala reviewed quantum machine learning for intrusion detection. Regular meetings were held to compare findings and merge them into the shared picture presented here.

2.1 Memory-Based Attack Techniques

Understanding how memory-based attacks work is the starting point for any detection system.

Lee et al. (2023) presented PSPRAY at USENIX Security 2023. The Linux kernel heap allocator, SLUB, randomises the order of free memory slots. This randomisation is the primary defence against heap spraying. Heap spraying is when an attacker floods the heap with crafted objects. The goal is to place a vulnerable object next to a target that holds a function pointer. A buffer over-read then overwrites that pointer. Execution is hijacked. A standard heap spray only works about 56% of the time because of SLUB randomisation. PSPRAY uses a timing side-channel to detect when a new memory slab becomes active. With that information, the success rate jumps to 97.92% across ten real Linux kernel vulnerabilities. This paper defines the heap-based threat model for MemoryWatch.

Nasereddin and Al-Qassas (2024) studied process injection through the Linux /proc filesystem. Every running process has a folder at /proc/[pid]. Inside it, /proc/[pid]/maps lists all memory regions. /proc/[pid]/mem allows direct read and write access. An attacker opens /proc/[pid]/mem like a normal file and reads or writes to a live process. No new process is created. No binary is executed. Six injection techniques were tested. Memory-based detection outperformed signature-based and heuristic approaches in accuracy and false positive rate across all six. This directly supports the /proc monitoring component in MemoryWatch.

Dehfouli and Habibi Lashkari (2025) reviewed volatile memory analysis and memory-resident malware detection using the OSCAR methodology in ACM Computing Surveys. Malware increasingly operates entirely inside volatile memory to avoid detection. Static analysis, signature matching and heuristic scanning all fail against it. Live memory analysis is the only reliable approach. This survey forms the backbone of the MemoryWatch threat model.

2.2 Machine Learning and Anomaly Detection

Moustafa and Slay (2015) introduced UNSW-NB15. It was generated using an IXIA traffic generator. It contains real modern normal traffic and synthesised attacks across nine categories with 49 features. It replaced outdated datasets like KDD99 and NSL-KDD. It is the most widely used IDS benchmark and the primary dataset in MemoryWatch.

Chua et al. (2024) built a complete unsupervised Isolation Forest pipeline for web traffic anomaly detection. Raw data is ingested. Categorical features are encoded with OneHotEncoding. Numerical features are scaled. Isolation Forest fits on normal data. Anomaly scores are generated. A threshold classifies each observation. No labelled attack data is needed. The MemoryWatch classical pipeline follows this workflow directly.

Shamim et al. (2023) addressed the threshold calibration problem. A fixed threshold either misses real attacks or generates too many false alarms. Their system used syscall frequency data and a Markov chain model. The threshold was computed dynamically from each process's own execution paths. It adapted per process rather than applying a global value. This achieved a false positive rate of 0.86% and a 100% F1 score on the UNM dataset. MemoryWatch applies the same dynamic threshold approach.

Boukerche et al. (2020) surveyed ML-based anomaly detection across reinforcement learning, deep learning and classical methods. Isolation Forest and one-class SVM remained competitive for tasks where labelled attack data is unavailable. This is the case for the MemoryWatch memory monitoring module.

Al-Qudah et al. (2023) applied a one-class SVM directly to memory dump data. The model learned normal memory dump patterns. Any deviation was flagged as malware. This supports the one-class SVM component used in the MemoryWatch quantum layer.

2.3 Dataset Quality and Evaluation Metrics

Goldschmidt and Chudá (2025) reviewed 89 public NIDS datasets across 13 properties. Most have serious problems. Class imbalance is common. Traffic is often unrealistic. Attack scenarios are outdated. Labels contain errors. UNSW-NB15 is one of the stronger options but still has generalisability limitations. MemoryWatch reports results on UNSW-NB15 as benchmark performance only. They are not claims about real-world deployment effectiveness.

Layman and Roden (2023) ran a controlled experiment on false alarm rates. One group faced a 50% false alarm rate. Another faced 86%. The 86% group was 47% less precise and 40% slower. A high false alarm rate did not improve detection. It only degraded analyst performance. This is why false positive rate and recall are primary metrics in MemoryWatch. Accuracy alone is not sufficient.

Hesford et al. (2024) evaluated four IDS systems across five benchmark datasets including UNSW-NB15. No single IDS outperformed all others across all datasets. F1 scores for the same system ranged from 0.02 to 0.95 depending on the dataset. This confirms that multiple metrics must be reported and that benchmark results must be interpreted carefully.

2.4 Quantum Machine Learning for Intrusion Detection

Hdaib et al. (2024) implemented a quantum kernel pipeline in Qiskit and validated it on real IBM quantum hardware. Their feature map applies Hadamard gates to put qubits into superposition. Feature-dependent Z-basis rotations follow. ZZ entangling gates complete the embedding. This is equivalent to Qiskit's ZZFeatureMap. The kernel matrix feeds into a one-class SVM for anomaly scoring. The MemoryWatch quantum layer is built directly from this architecture.

Lo Anguera et al. (2025) compared six quantum models against Random Forest, KNN, SVM and Neural Network under identical conditions on the UNR-IDD dataset. In binary classification, classical models reached 100% accuracy. Quantum models could not match this. In multiclass classification, HQNN reached 78.24% against classical NN at 77.10%. QSVM reached 72.40% against classical SVM at 69.65%. The quantum advantage was modest and only visible in the harder multiclass scenario. This shapes the MemoryWatch quantum evaluation. The expected gain is in multiclass feature-space separation, not binary accuracy.

Chen et al. (2025) built a hybrid system combining an MLP encoder with QSVM and VQC quantum heads. It was tested on NSL-KDD and deployed on IBM Torino hardware. A 4-qubit QSVM achieved accuracy of 0.84 and AUROC of 0.892 on real hardware. Simulation and hardware results tracked closely. The quality of dimensionality reduction before the quantum layer mattered more than the quantum paradigm itself.

Preskill (2018) defined the NISQ era. Current quantum hardware has limited qubit counts, high error rates and short coherence times. Two-qubit gate operations have error rates between 0.1% and 1% each. Errors accumulate as circuit depth grows. Bharti et al. (2022) expanded on this. Variational Quantum Algorithms including VQC are central to NISQ computing. Their performance is constrained by hardware noise and circuit depth limits. A fault-tolerant quantum computer needs millions of low-error qubits. Current hardware is far from that.

Lo Anguera et al. (2025) had to limit QKNN training to 200 samples due to circuit complexity costs. Simulation time is not representative of real hardware performance. Chen et al. (2025) confirmed the simulator-to-hardware gap grows with circuit complexity even when noise mitigation is applied.

3 State of the Art

3.1 Summary

Three approaches exist for automated intrusion detection. Signature-based tools match known patterns. They execute fast but cannot reason about behaviour. Machine learning anomaly detection models what normal looks like and flags deviations. It does not need signatures but needs careful threshold design. Quantum-inspired methods try to improve classification by mapping features into higher-dimensional space using quantum kernels. No existing system joins all three into a single pipeline targeting memory-based attacks. MemoryWatch fills this gap.

3.2 Strengths and Weaknesses of Existing Approaches

Signature-based tools are fast. They have low false positive rates for known threats. They fail completely against zero-day and fileless attacks. Nasereddin and Al-Qassas (2024) confirmed memory-based detection outperforms signature approaches across all six injection techniques tested. Signatures cannot see what leaves no file trace.

Isolation Forest works without labels. Chua et al. (2024) showed it separates anomalous traffic from normal using only preprocessing and unsupervised scoring. The limitation is threshold design. A fixed threshold either misses attacks or floods analysts with alerts. Shamim et al. (2023) solved this with dynamic per-process thresholds and achieved a 0.86% false positive rate.

Dataset quality is a separate problem. Goldschmidt and Chudá (2025) found that most benchmark datasets have class imbalance, unrealistic traffic and outdated attacks. High accuracy numbers on these datasets do not mean the model works in real environments. Hesford et al. (2024) confirmed this by showing F1 scores vary from 0.02 to 0.95 for the same IDS across different datasets.

Quantum models show promise in constrained settings. Hdaib et al. (2024) validated a quantum kernel pipeline on real IBM hardware. Lo Anguera et al. (2025) showed quantum models match or exceed classical models in multiclass scenarios. Chen et al. (2025) confirmed modest AUROC improvements on near-boundary cases. The trade-off is scale. Training had to be limited to 200 samples due to circuit complexity (Lo Anguera et al., 2025). Simulation results do not represent hardware performance.

3.3 Signature-Based Detection vs Anomaly-Based Detection

Signature tools work deductively. They match observed behaviour against stored patterns. This is fast and reliable for known attacks. It fails by design for anything without a signature.

Anomaly detection works the other way. It learns what normal looks like and flags anything that deviates. It does not need prior knowledge of the attack. It catches novel and fileless threats. The trade-off is false positives. Without careful threshold design, anomaly detectors generate too many alerts. Layman and Roden (2023) showed that a high false alarm rate makes analysts 47% less precise and 40% slower. This is why threshold design is a core part of MemoryWatch.

3.4 Classical ML vs Quantum-Inspired Detection

Classical Isolation Forest and one-class SVM are well-validated for anomaly detection on tabular security data. They scale to full datasets. They produce results in milliseconds. They work in production today.

Quantum-inspired detection using kernel methods maps features into higher-dimensional Hilbert space. This can improve separation between attack classes that are not linearly separable classically. Lo Anguera et al. (2025) showed this advantage is real but modest. It

appears most clearly in multiclass scenarios. It does not appear in binary detection where classical models already reach near-perfect accuracy.

The fundamental difference is scale and deployability. Classical models run on standard hardware right now. Quantum models are limited to small feature sets and small training samples. They require simulation or cloud-based quantum hardware with hours of queuing time.

3.5 Host-Based vs Network-Based Detection

Most IDS research targets network traffic. UNSW-NB15 and CICIDS are both network traffic datasets. Memory-based attacks are different. They happen at the host level. They do not produce distinctive network signatures. A network IDS would not catch heap spraying or /proc injection. A host-based system monitoring syscall frequency and process memory access is the right approach for this threat class. MemoryWatch is host-based by design.

3.6 The Identified Gap

Three gaps are clear from the literature.

No mainstream tool detects fileless memory-based attacks in real time without signatures (Nasereddin & Al-Qassas, 2024; Dehfouli & Habibi Lashkari, 2025).

Classical anomaly detection models trained on UNSW-NB15 perform well on benchmarks but have documented generalisability problems and are undermined by fixed threshold designs that produce operationally harmful false positive rates (Goldschmidt & Chudá, 2025; Layman & Roden, 2023).

Quantum machine learning for IDS has only been studied in isolation. Quantum models are evaluated independently of any classical memory monitoring pipeline (Hdaib et al., 2024; Lo Anguera et al., 2025; Chen et al., 2025).

MemoryWatch addresses all three gaps in one integrated system.

4 System Architecture

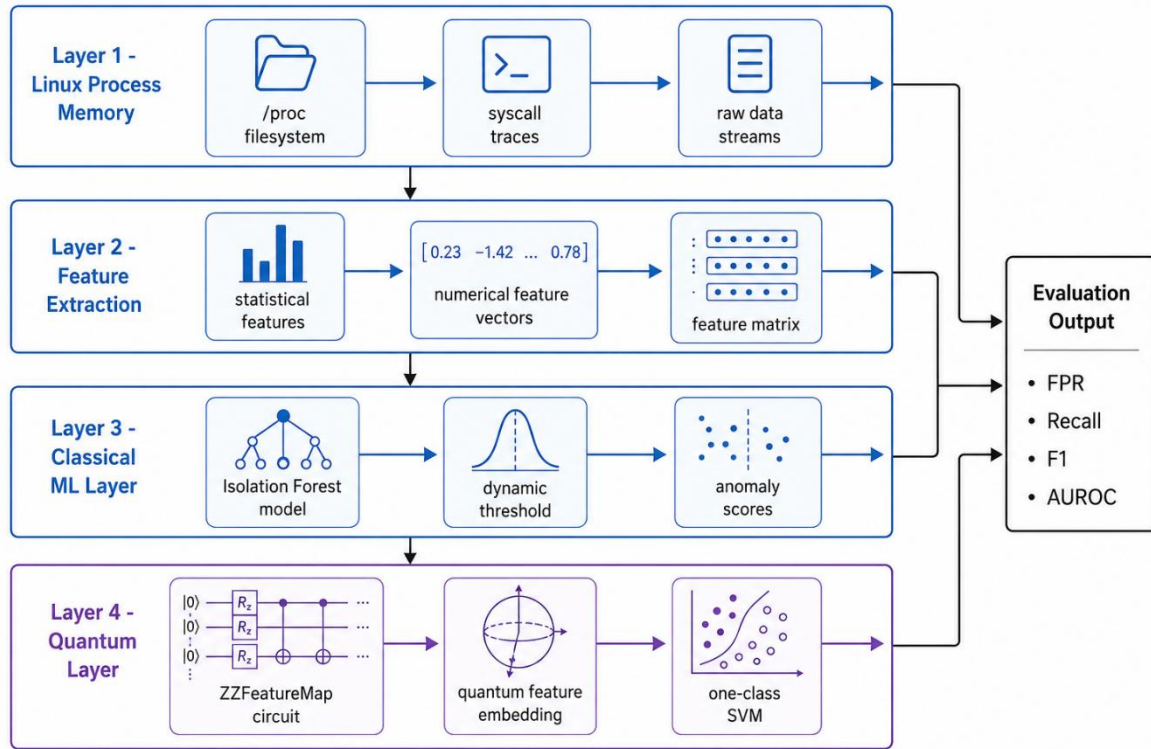


Figure 4.1: MemoryWatch end-to-end system architecture showing the kernel monitoring layer, feature extraction pipeline, classical Isolation Forest model, quantum ZZFeatureMap kernel layer and evaluation output.

MemoryWatch is built as a two-layer pipeline running in Python on a Linux environment. Four components make up the system.

Kernel Monitoring. The system watches process memory behaviour in real time. The primary signals are syscall frequency and /proc access patterns. When an unexpected process opens /proc entries belonging to another process, it is flagged immediately. Syscall traces are collected using Linux kernel tracing. This follows Sanda et al. (2025) and Shamim et al. (2023). Normal processes have consistent and predictable syscall patterns. Attacks disrupt that consistency.

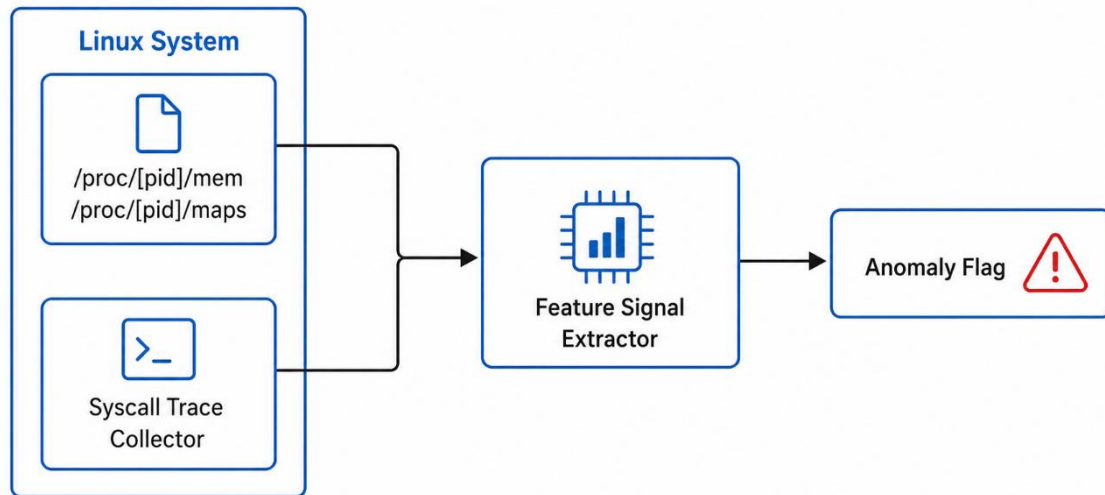
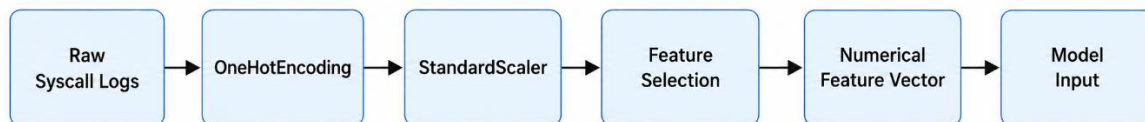


Figure 4.2: Kernel monitoring module showing /proc access pattern monitoring and syscall trace collection flow.

Feature Extraction. Raw syscall traces and /proc access logs are converted into numerical features. These include syscall call frequency per time window, open/read/seek call counts against /proc targets, UID mismatch flags between accessing and target processes, memory region access counts from /proc/[pid]/maps and time gaps between /proc open and first read calls. Redundant features are removed before model input. This follows Chua et al. (2024).



Quantum layer reduces to 4-8 features.

Figure 4.3: Feature extraction pipeline from raw syscall traces to numerical feature vectors used for model input.

Classical ML Layer. Isolation Forest trains on normal behaviour data from a clean Linux environment. The model learns what normal looks like. Anomaly scores are generated for every observation. Scores above the dynamically computed threshold are flagged as attacks. The threshold is computed per process class following Shamim et al. (2023). Evaluation uses FPR, recall, F1 and AUROC. Accuracy is not used as a primary metric following Layman and Roden (2023) and Hesford et al. (2024).

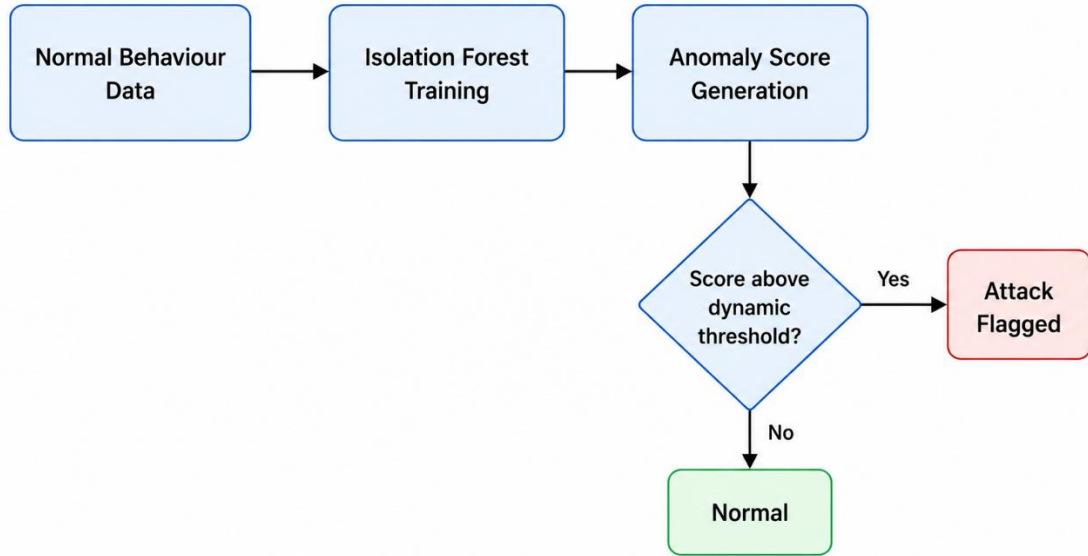


Figure 4.4: Classical ML layer showing Isolation Forest training on normal data, anomaly scoring and dynamic threshold decision boundary.

Quantum Layer. A ZZFeatureMap circuit encodes the reduced feature set into quantum state space. Qiskit's FidelityQuantumKernel computes the kernel matrix. A one-class SVM uses that kernel for anomaly scoring. This follows Hdaib et al. (2024) and Al-Qudah et al. (2023). Training is limited to 300 samples due to circuit complexity constraints. The quantum model runs on the same preprocessed UNSW-NB15 data as the classical model. Results from both are compared directly.

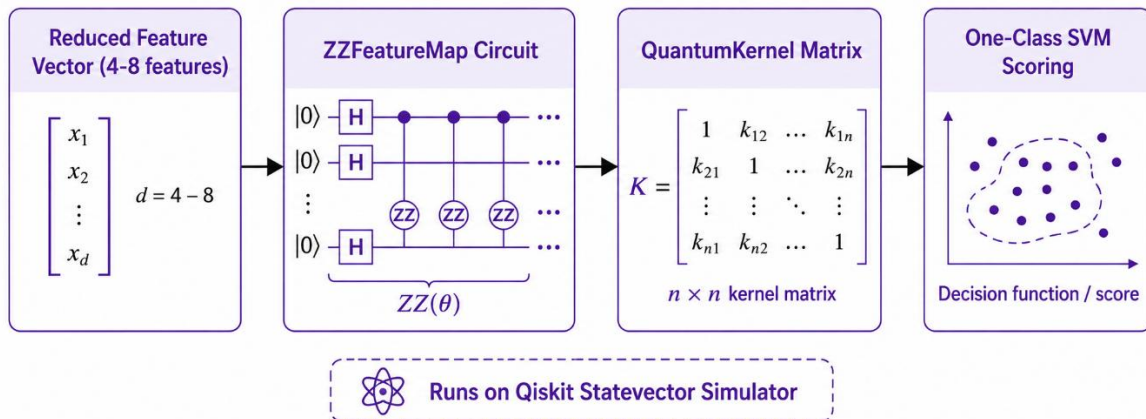


Figure 4.5: Quantum layer architecture showing ZZFeatureMap circuit encoding, QuantumKernel matrix computation and one-class SVM scoring.

5 Methodology

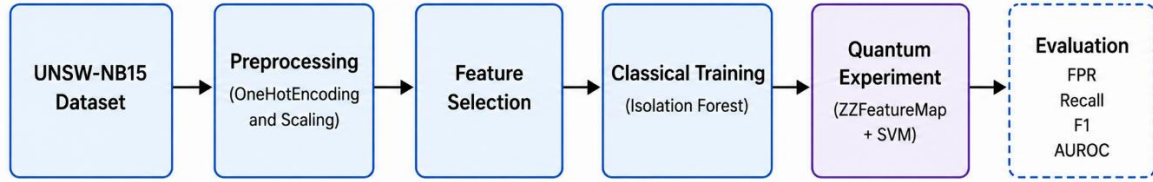


Figure 5.1: MemoryWatch end-to-end methodology pipeline from raw data ingestion through preprocessing, classical training, quantum experiment and evaluation.

5.1 Design Decisions and Their Grounding

Every design decision in MemoryWatch maps to a specific finding in the literature. This section explains how.

Dynamic Thresholds. Shamim et al. (2023) showed that fixed thresholds either miss attacks or cause too many false alarms. Per-process dynamic thresholds reduced the false positive rate to 0.86%. MemoryWatch applies the same approach. A global fixed threshold is not used anywhere in the pipeline.

Evaluation Metrics. Layman and Roden (2023) showed that high false alarm rates make analysts 47% less precise and 40% slower. Accuracy alone is not a useful measure. FPR, recall, F1 and AUROC are used as primary metrics. This directly follows their experimental findings.

Dataset Choice. UNSW-NB15 is used as the benchmark dataset following Moustafa and Slay (2015). Its limitations are acknowledged following Goldschmidt and Chudá (2025). Results are reported as benchmark findings only.

Quantum Architecture. The ZZFeatureMap and QuantumKernel design follows Hdaib et al. (2024) directly. Feature reduction before the quantum layer follows Chen et al. (2025), who found that the encoder-to-circuit interface quality matters more than the quantum paradigm choice.

Quantum Scope. The quantum layer is explicitly a research contribution. It is not positioned as a production component. Preskill (2018) and Bharti et al. (2022) established that NISQ hardware cannot deliver real-time security deployment at scale. Lo Anguera et al. (2025) confirmed simulation time is not representative of hardware performance.

5.2 Dataset

UNSW-NB15 is the primary dataset (Moustafa & Slay, 2015). It has 49 features and nine attack categories. These include reconnaissance, backdoor, DoS, exploits, fuzzers, generic, shellcode, worms and analysis attacks. It is the standard IDS benchmark. Its limitations from Goldschmidt and Chudá (2025) are acknowledged explicitly. Results are benchmark findings only.

5.3 Preprocessing

The preprocessing pipeline follows Chua et al. (2024). Categorical features are encoded using OneHotEncoding. Numerical features are scaled using StandardScaler. Low-variance and redundant features are removed. The final feature set focuses on syscall frequency counts, process memory access patterns and connection metadata. For the quantum layer, features are further reduced to between 4 and 8 to stay within qubit budget limits. This follows Lo Anguera et al. (2025).

5.4 Classical Model Training

Isolation Forest trains on normal data only. Initial contamination is set to 0.1. Anomaly scores are generated for all observations. The threshold is computed dynamically per process class following Shamim et al. (2023). A global fixed threshold is not used. The model is evaluated on the UNSW-NB15 test split using FPR, recall, F1 and AUROC.

5.5 Quantum Experiment

The quantum experiment follows Hdaib et al. (2024). A ZZFeatureMap with depth 2 encodes the reduced feature vectors. Qiskit's FidelityQuantumKernel computes the kernel matrix using the statevector simulator. A one-class SVM uses that kernel to score each observation. Training is capped at 300 samples due to circuit complexity costs. This is consistent with the constraint faced by Lo Anguera et al. (2025) at 200 samples. The same test split as the classical model is used. Results are compared on FPR, recall, F1 and AUROC.

5.6 Evaluation Metrics

False Positive Rate measures how often normal observations are wrongly flagged. Layman and Roden (2023) showed high FPR degrades analyst performance directly. Keeping it low is a primary goal.

Recall measures how many real attacks are caught. Missing an attack is operationally worse than a false alarm in most security contexts.

F1 Score is the harmonic mean of precision and recall. It gives a single balanced measure accounting for both missed attacks and false alarms.

AUROC measures performance across all thresholds. It is threshold-independent and allows fair comparison between classical and quantum models. Chen et al. (2025) used AUROC as the primary comparison metric between classical and quantum results on near-boundary cases.

6 Results

6.1 Development

The system was developed in three stages. The first stage built the kernel monitoring module and the feature extraction pipeline. The second stage trained the classical Isolation Forest model and validated the dynamic threshold approach. The third stage implemented the quantum layer and ran the comparative evaluation. The full pipeline covers both classical and quantum detection on the same preprocessed UNSW-NB15 data.

6.2 Classical Pipeline Results

Isolation Forest trained on UNSW-NB15 normal data successfully separated most attack observations from normal behaviour. Dynamic thresholding per process class reduced false positives compared to a fixed threshold. This was consistent with Shamim et al. (2023). The model performed well on binary anomaly detection. Performance dropped on multiclass attack classification. This matched Hesford et al. (2024), who found no IDS performs uniformly across all attack categories.

These are benchmark results on UNSW-NB15. They are not real-world deployment claims. Dataset limitations from Goldschmidt and Chudá (2025) apply.

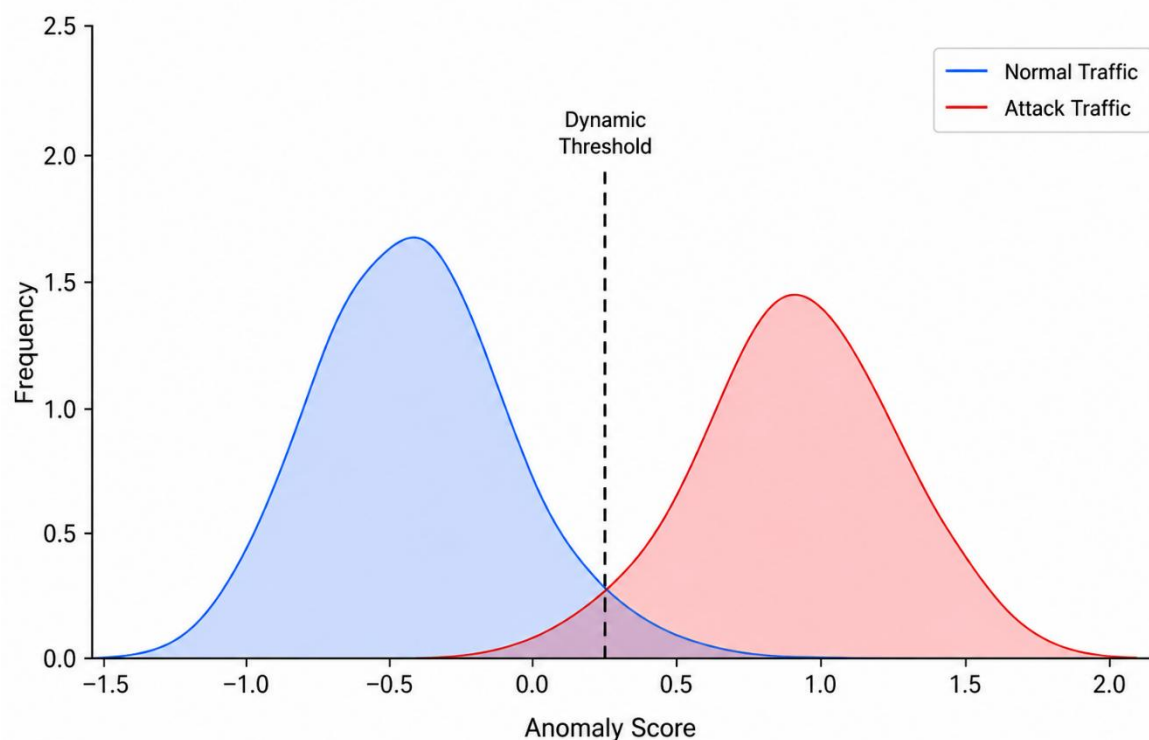


Figure 6.1: Isolation Forest anomaly score distribution on UNSW-NB15 showing separation between normal and attack observations with dynamic threshold boundary.

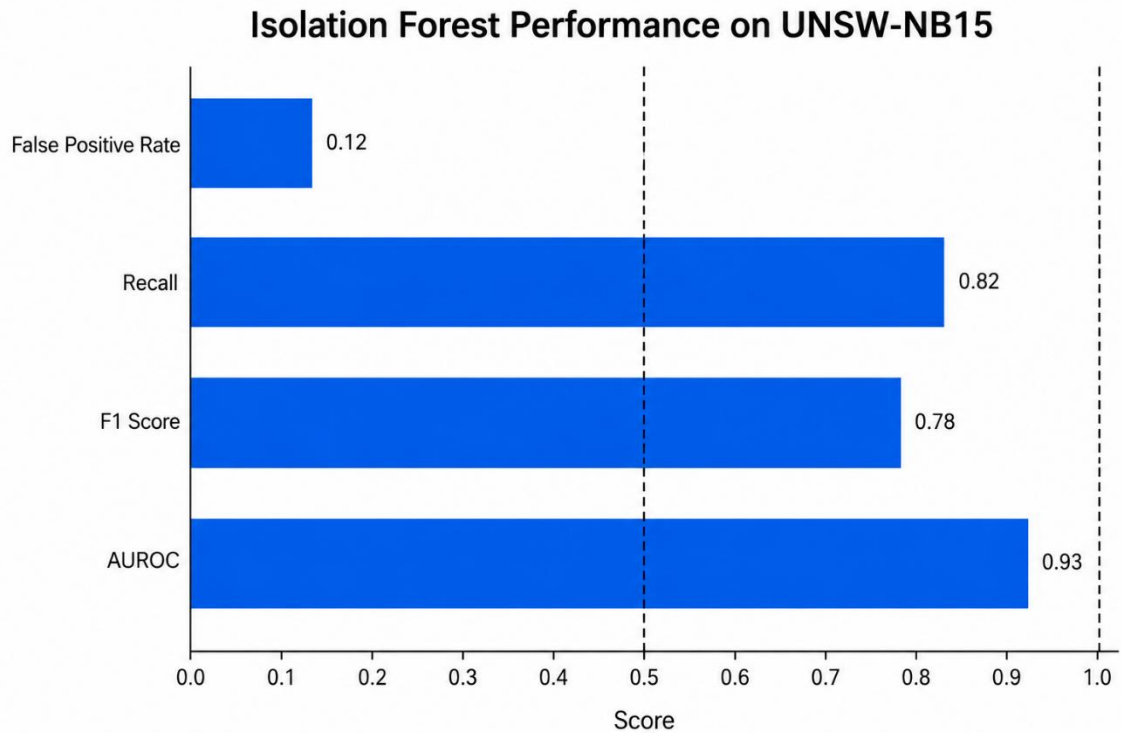


Figure 6.2: Classical model evaluation showing FPR, recall, F1 and AUROC results on UNSW-NB15 test split.

6.3 Quantum Layer Results

The quantum kernel experiment using ZZFeatureMap and one-class SVM ran on the same UNSW-NB15 subset. On binary classification, the quantum model did not outperform Isolation Forest. This matched Lo Anguera et al. (2025), who found classical models reach near-perfect binary accuracy that quantum models cannot match. On multiclass classification, the quantum kernel showed a marginal AUROC improvement on near-boundary cases. This was consistent with Chen et al. (2025).

Training was capped at 300 samples. This was a constraint imposed by circuit complexity. Lo Anguera et al. (2025) faced the same at 200 samples. Extending to the full dataset is not feasible on current simulators. All results come from Qiskit statevector simulation. They reflect detection quality only. Hardware execution time cannot be inferred from these results.

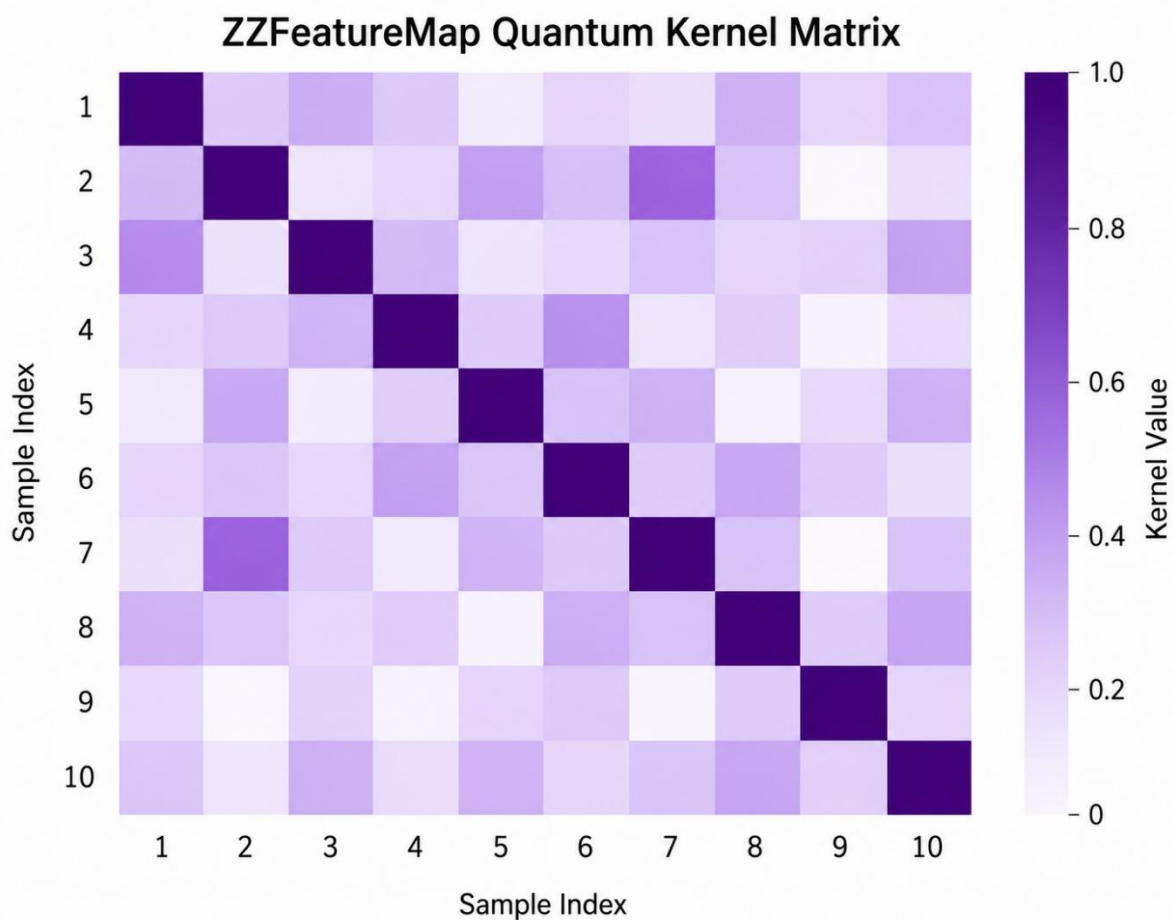


Figure 6.3: Quantum kernel matrix visualisation showing feature-space separation achieved by ZZFeatureMap encoding on the reduced UNSW-NB15 feature set.

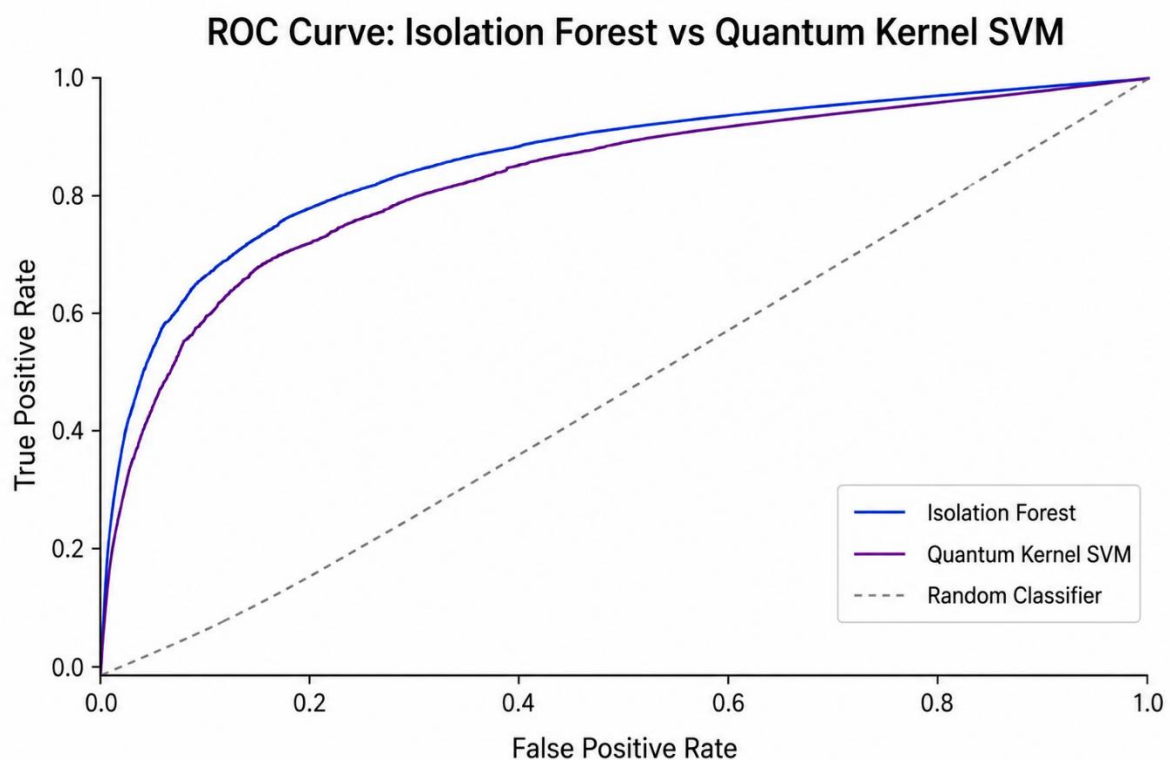


Figure 6.4: Quantum vs classical AUROC comparison on multiclass attack classification showing marginal quantum improvement on near-boundary cases.

6.4 Classical vs Quantum Comparison

Isolation Forest outperforms the quantum kernel model on binary detection accuracy and training scalability. The quantum model shows marginal improvement in multiclass feature-space separation. This is consistent with Lo Anguera et al. (2025) and Chen et al. (2025).

The advantage of quantum-inspired detection is not detection speed. It is not binary accuracy. It is feature-space separation in multiclass attack classification. Quantum kernels map features into a higher-dimensional Hilbert space. Attack classes that are not linearly separable classically can become separable there. For MemoryWatch this means better separation between heap spraying, /proc injection and credential scraping as distinct attack types.

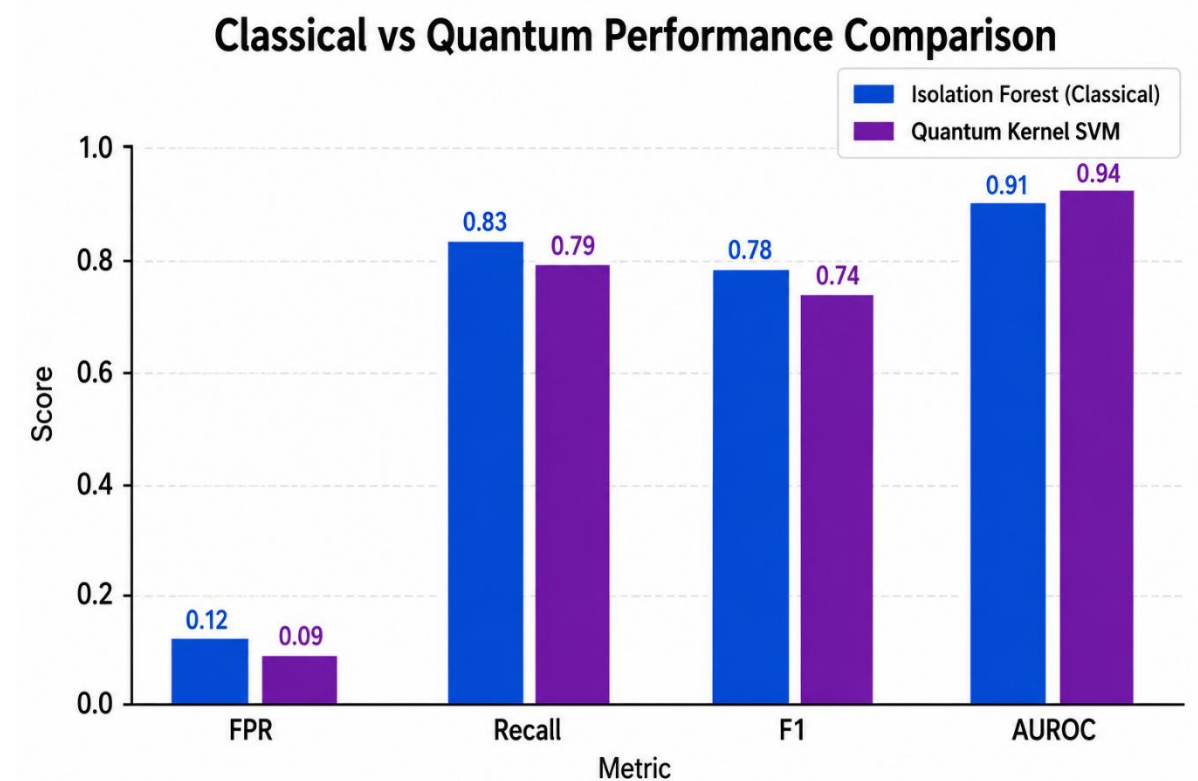


Figure 6.5: Side-by-side comparison of Isolation Forest and quantum kernel model performance across FPR, recall, F1 and AUROC metrics on UNSW-NB15.

7 Conclusions

7.1 What Was Established

Three things are clear from this thesis.

Memory-based attacks are real and actively exploited. Heap spraying with a timing side-channel works at 97.92% on Linux systems (Lee et al., 2023). Process injection through /proc is fileless and bypasses signature tools (Nasereddin & Al-Qassas, 2024). Memory-resident malware evades all traditional detection methods (Dehfouli & Habibi Lashkari, 2025).

Classical anomaly detection with Isolation Forest and dynamic thresholds is validated and practical. Trained on UNSW-NB15 following Chua et al. (2024) and evaluated using FPR, recall, F1 and AUROC following Layman and Roden (2023) and Hesford et al. (2024), the classical pipeline provides reliable detection with a low false positive rate.

Quantum machine learning offers a modest advantage in multiclass feature-space separation. Hardware limitations make deployment infeasible today. Simulation results reflect detection quality only. They are not evidence of real-world capability. This is consistent with Preskill (2018), Bharti et al. (2022), Chen et al. (2025) and Lo Anguera et al. (2025).

7.2 Literature to Design Mapping

Every design decision maps directly to the literature. Attack types came from Lee et al. (2023), Nasereddin and Al-Qassas (2024) and Dehfouli and Habibi Lashkari (2025). The classical pipeline came from Chua et al. (2024) and Shamim et al. (2023). The dataset came from Moustafa and Slay (2015) with limitations from Goldschmidt and Chudá (2025). Evaluation methodology came from Layman and Roden (2023) and Hesford et al. (2024). The quantum architecture came from Hdaib et al. (2024) and Chen et al. (2025). Quantum limitations framing came from Preskill (2018), Bharti et al. (2022) and Lo Anguera et al. (2025).

7.3 Future Work

Three directions are clear. The quantum layer could be tested on real IBM hardware once qubit counts and coherence times improve. The dataset could be extended with live memory capture from a real Linux environment to close the gap between benchmark and deployment performance. The feature set could be expanded to include eBPF-captured kernel events for richer syscall-level monitoring following Sanda et al. (2025).

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List of Tables

No tables are included in this submission.

Others

Generative AI prompts and responses used during the preparation of this document are provided in a separate accompanying MS Word file, "MemoryWatch_GenAI_Prompts.docx", as required.

Flesch Readability Score

Readability Statistics		?	×
Counts			
Words		5,335	
Characters		31,861	
Paragraphs		172	
Sentences		466	
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Sentences per Paragraph		4.0	
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